



Corfu: A Cloud Scale Consistency Platform

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Code Mesh 2016

How a Typical Application Becomes Distributed



Questions

Search

how to scale an application

search

9,110 results

relevance

newest

votes

active

How a Typical Application Becomes Distributed



How a Typical Application Becomes Distributed



How a Typical Application Becomes Distributed



Questions

Search

build a distributed system from scratch

search

31 results

relevance

newest

votes

active

1

vote

0

answers

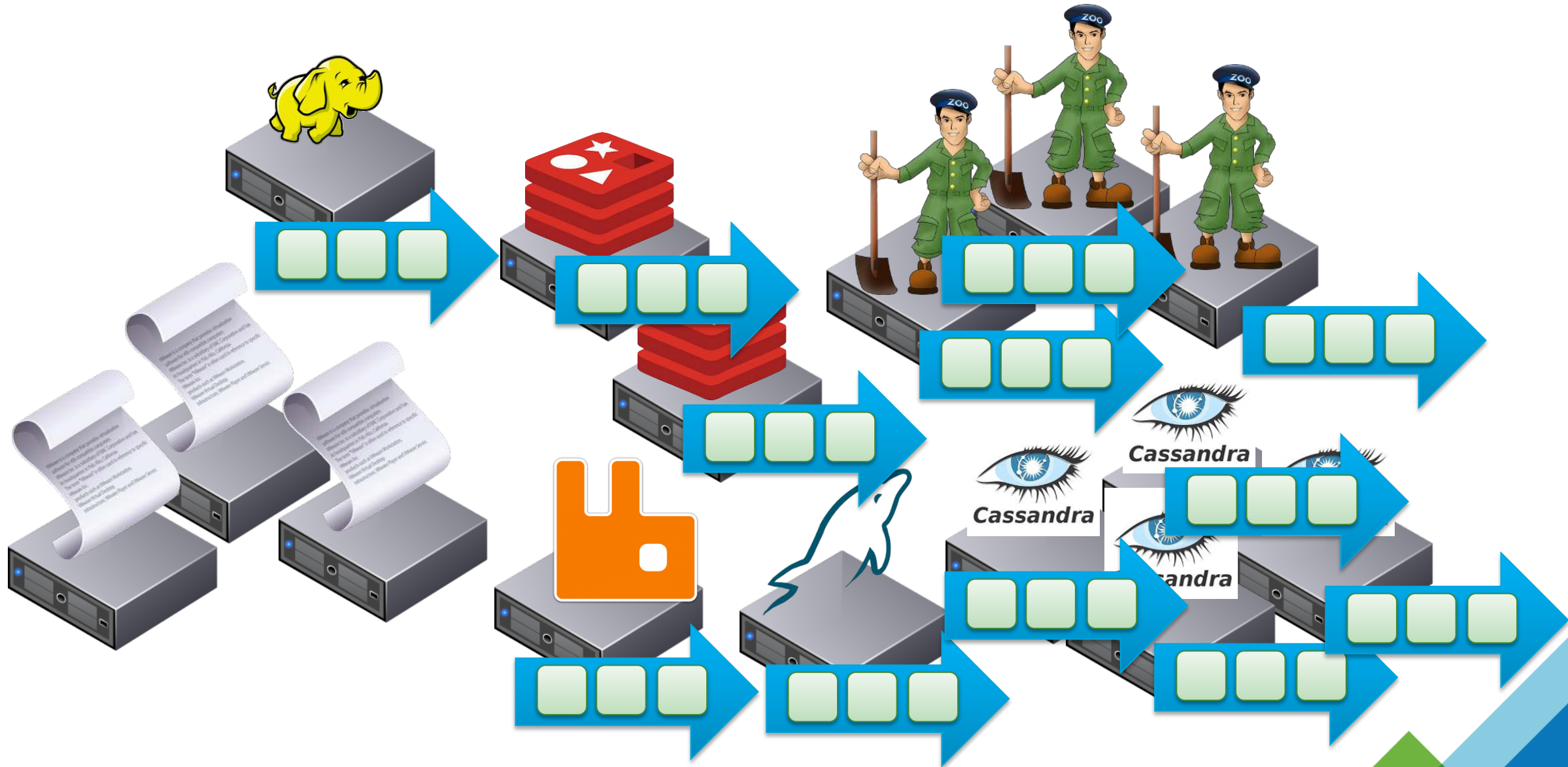
Q: What would a clean slate design of a distributed system look like?

Why would you ever want to do such a thing?

distributed-computing

distributed-system

How a Typical Application Becomes Distributed



What is Corfu?

Corfu



What is Corfu?

Corfu

Paxos



Paxos – A Family of Consensus Protocols

$A = 0$



$A = 0;$

Paxos – A Family of Consensus Protocols

$A = 1$



$A = 0;$
 $A = 1;$

Paxos – A Family of Consensus Protocols

$A = 1$



$A = 1$



$A = 1$



$A = 0;$
 $A = 1;$

Paxos – A Family of Consensus Protocols

$A = 2$



$A = 1$



$A = 1$



$A = 0;$
 $A = 1;$
 $A = 2;$

Paxos – A Family of Consensus Protocols

$A = 2$

$A = 2$

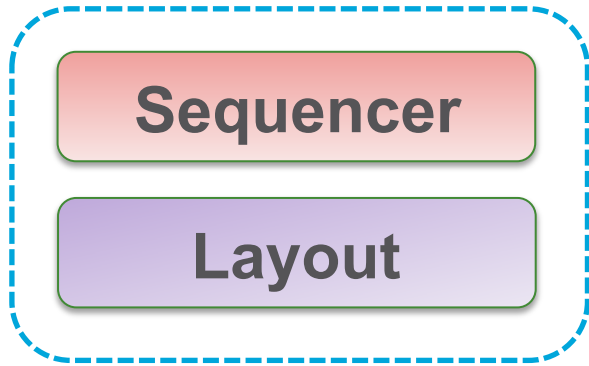


A Distributed Shared Log

```
A = 0;  
A = 1;  
A = 2;
```

10K ops/s
Doesn't Scale!

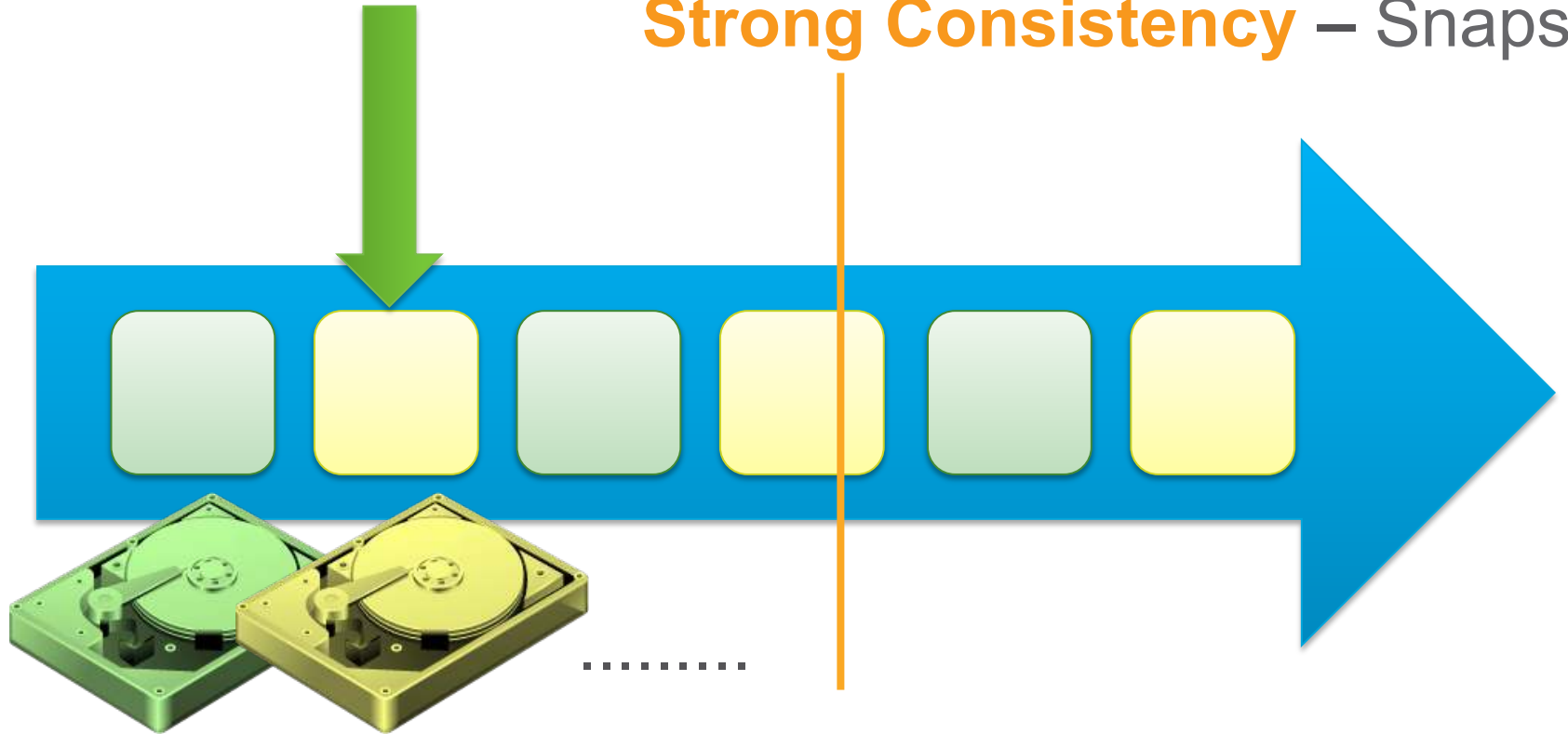
Corfu: A Distributed Shared Log



“Control Plane”
Traditional I/O
bottleneck

Fast Datapath

Strong Consistency – Snapshots



Write Scaling – Fast Updates

Read Scaling – Fast Random Reads

Corfu: A Distributed Shared Log

Sequencer

Layout

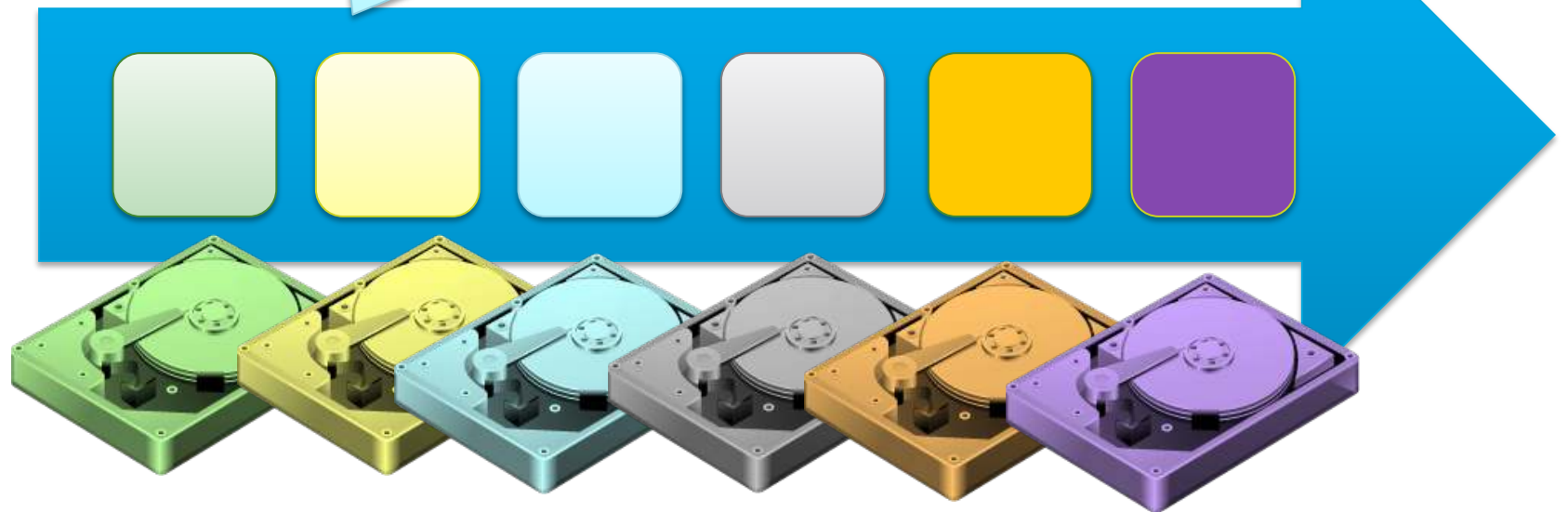
Sequencer is fast
600K op/s – not part of I/O path

Corfu: A Distributed Shared Log

Sequencer

Layout

Scalable
Add more stripes...



Corfu: A Distributed Shared Log

Sequencer

Layout

Fault Tolerant
Via chain replication...



F+1 Replicas
3 Replicas = 2 failures “per chain”

Corfu: A Distributed Shared Log

Sequencer

Layout

Strong Consistency
The log ... is the truth!

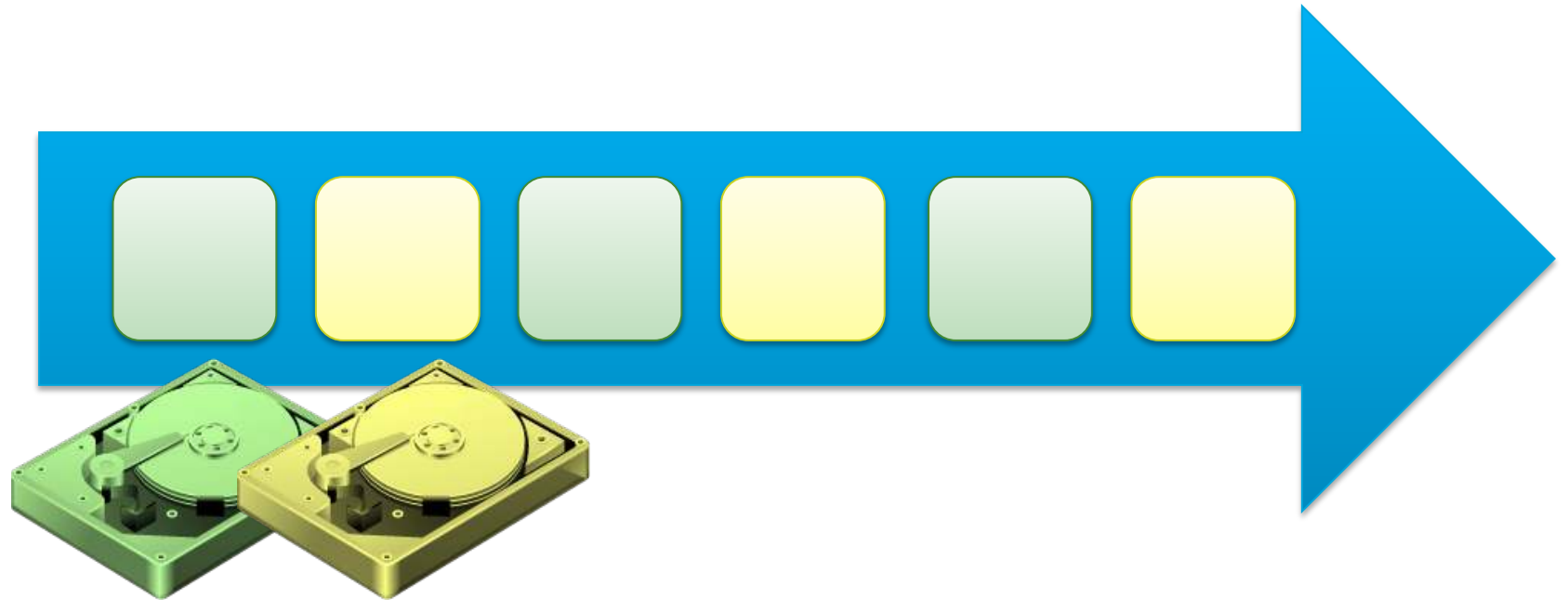


Corfu: A Distributed Shared Log

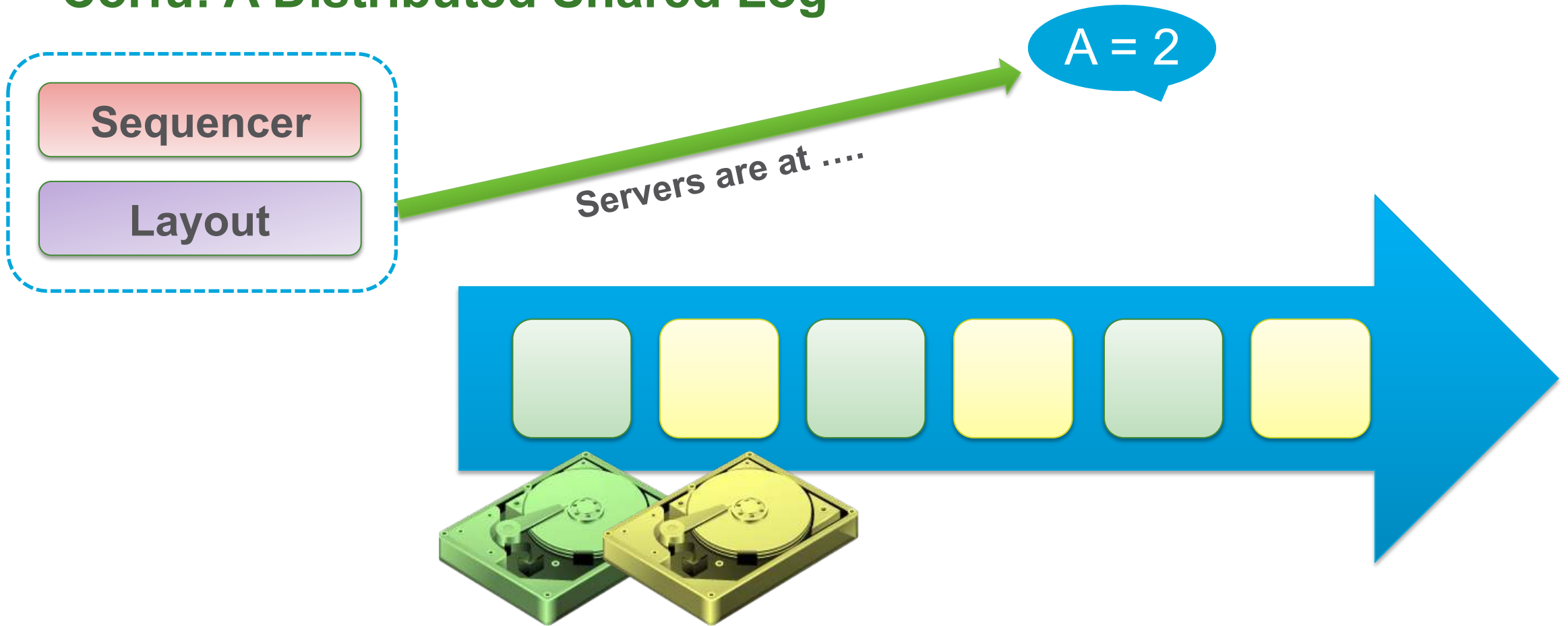
$A = 2$

Sequencer

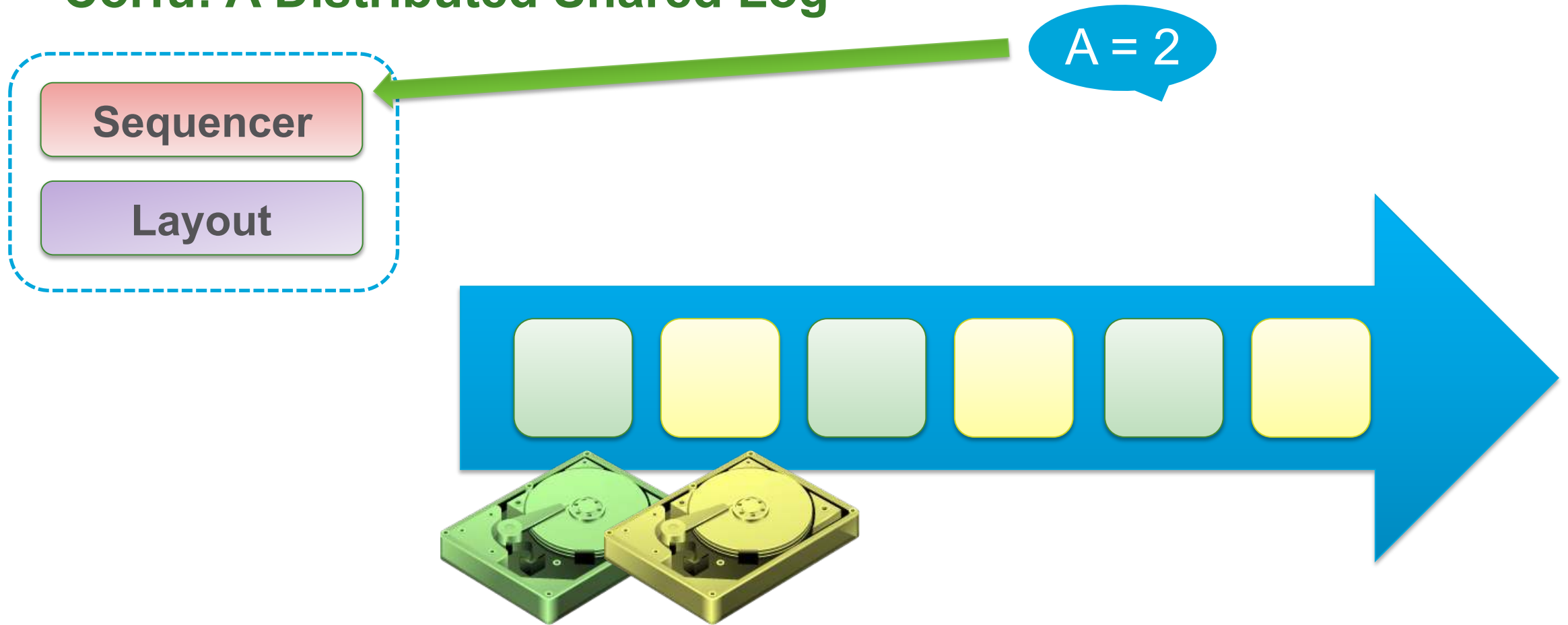
Layout



Corfu: A Distributed Shared Log



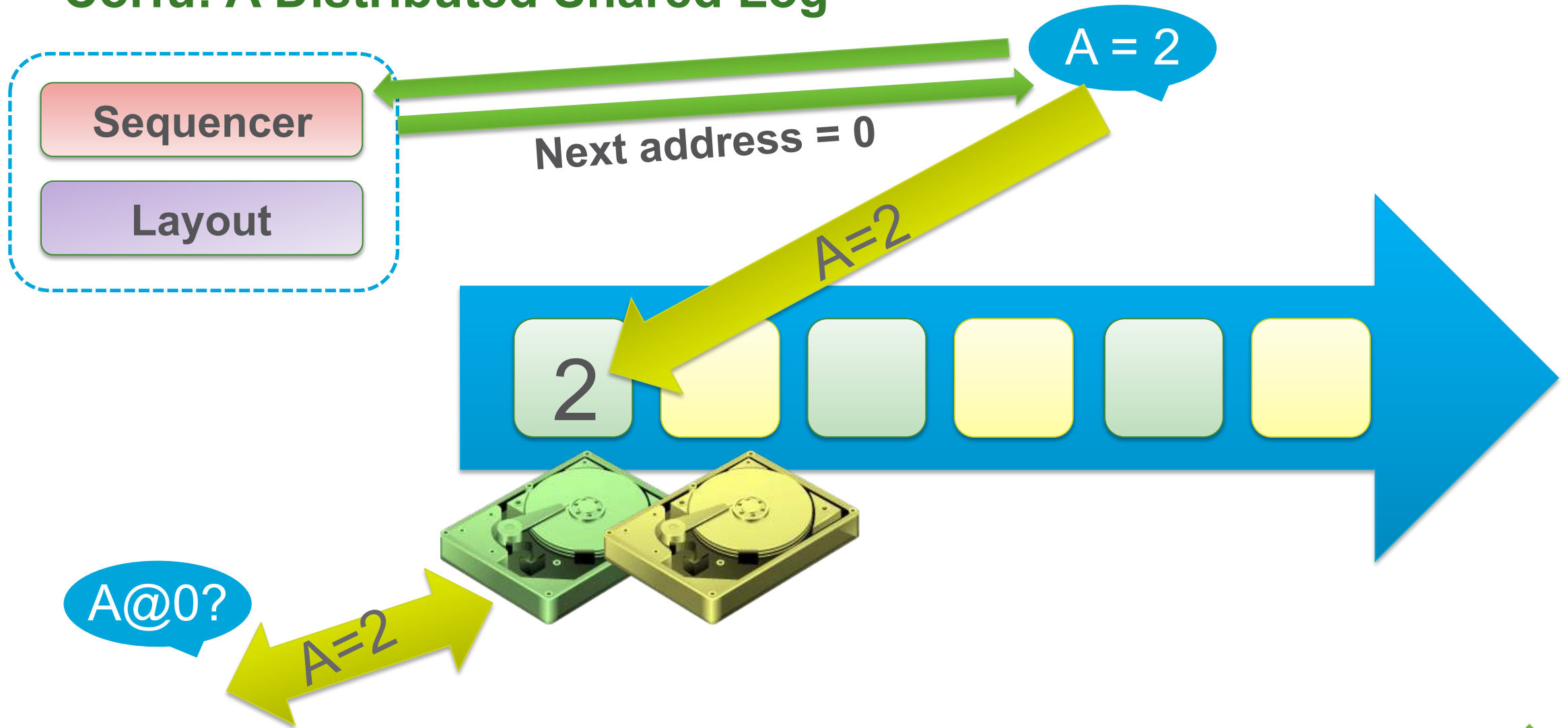
Corfu: A Distributed Shared Log



Corfu: A Distributed Shared Log



Corfu: A Distributed Shared Log



Corfu: A Distributed Shared Log



Corfu: A Distributed Shared Log



Corfu: A Distributed Shared Log

A
6

Sequencer

Layout

A=2

A=6

Corfu: A Distributed Shared Log

Sequencer

Layout

A
6

B

A=2

A=6

Corfu: A Distributed Shared Log

A@1 B@2

Sequencer

Layout

A
6

B
2

A=2

A=6

B=2

Corfu: A Distributed Shared Log

A@1 B@2

Sequencer

Layout

Transaction

A = 1;
B = 9;

A
6

B
2

A=2

A=6

B=2

Corfu: A Distributed Shared Log

A@3 B@3

Sequencer

Layout

Transaction

A = 1;
B = 9;

A
1

B
9

A=2

A=6

B=2

A=1
B=9

Corfu: A Distributed Shared Log

A@3 B@3

Sequencer

Layout

Transaction

```
if (A == 1) {  
  B = 7; }
```

A
1

B
9

A=2

A=6

B=2

A=1
B=9

Corfu: A Distributed Shared Log

A@3 B@3

Sequencer

Layout

Transaction

```
if (A == 1) {  
  B = 7; }
```

A
1

B
9

RS = A @ 3
WS = { B = 7 }

A=2

A=6

B=2

A=1
B=9

Address = 3

Corfu: A Distributed Shared Log

A@3 B@4

Sequencer

Layout

Transaction

```
if (A == 1) {  
  B = 7; }
```

A
1

B
7

RS = A @ 3
WS = { B = 7 }

A=2

A=6

B=2

A=1
B=9

B=7

Address = 3

Corfu: A Distributed Shared Log

A@4 B@3

Sequencer

Layout

Transaction

```
if (A == 1) {  
  B = 7; }
```

A
1

B
7

RS = A @ 3
WS = { B = 7 }

A=2

A=6

B=2

A=1
B=9

A=8

Address = 3

Corfu: A Distributed Shared Log

A@4 B@3

Sequencer

Layout

Transaction

```
if (A == 1) {  
  B = 7; }
```

A
1

B
7

~~R = A @ 3
W = { B = 7 }~~

A=2

A=6

B=2

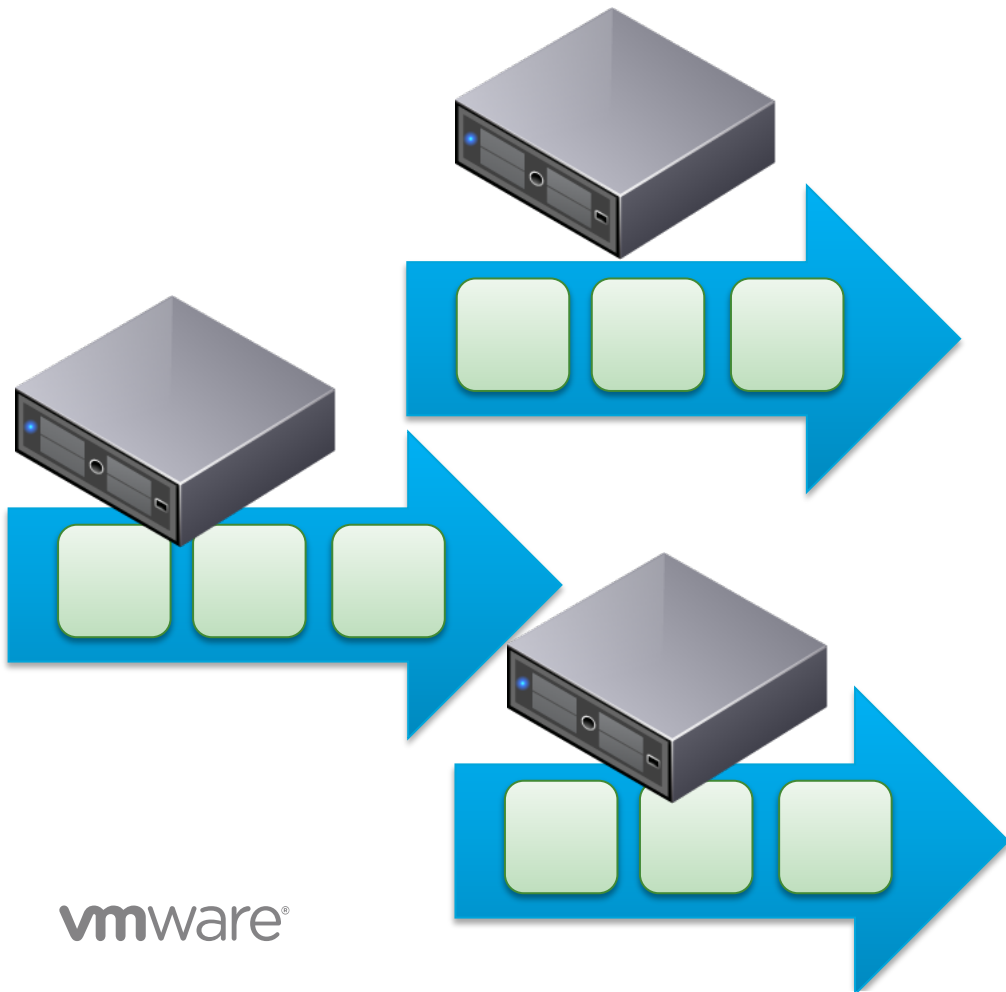
A=1
B=9

A=8

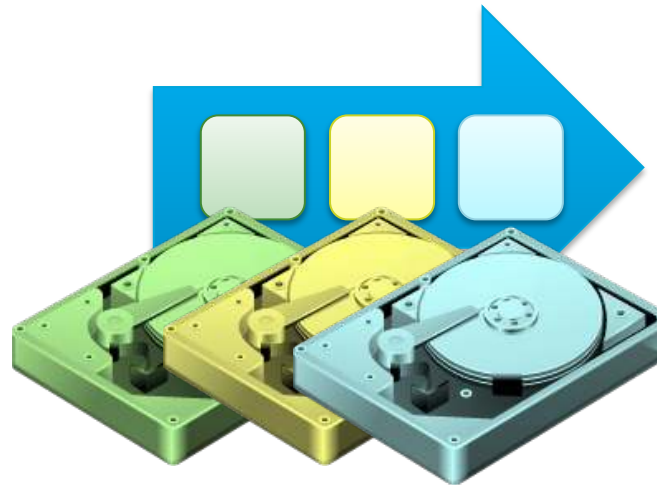
Address = 3

Corfu vs other “Distributed Logs”

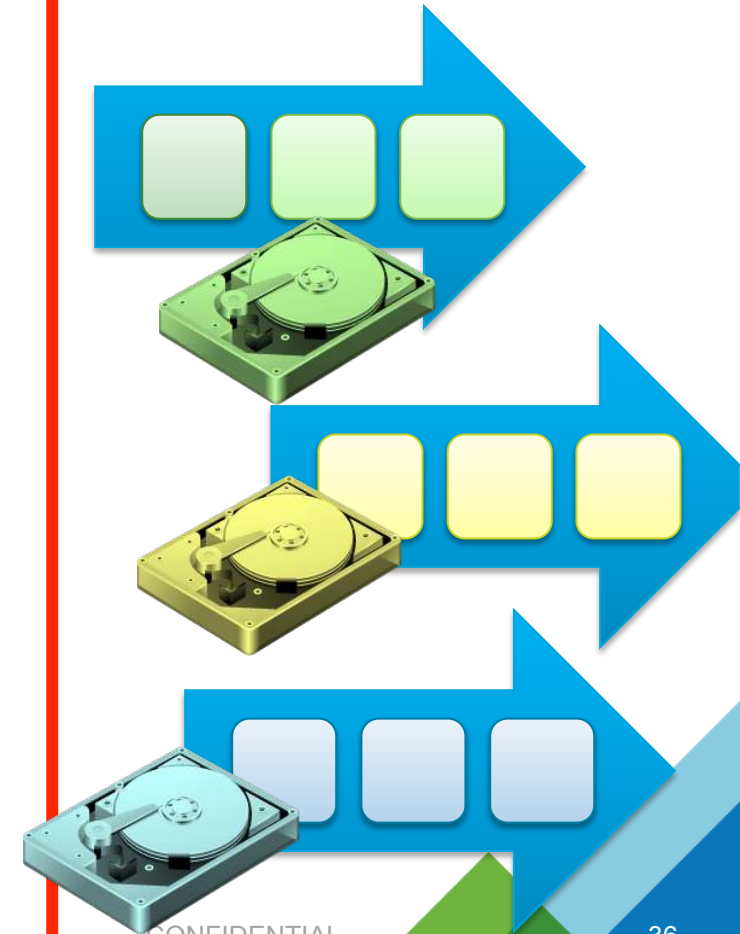
Paxos



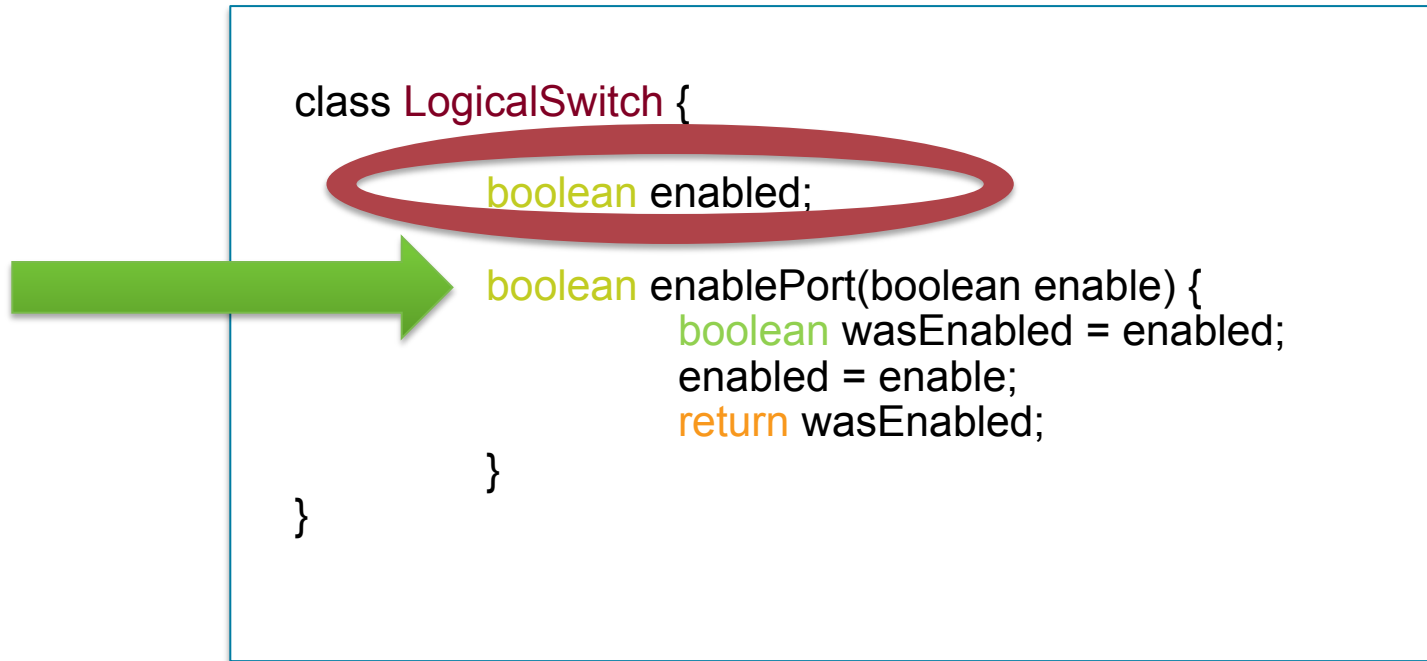
Corfu



Kafka



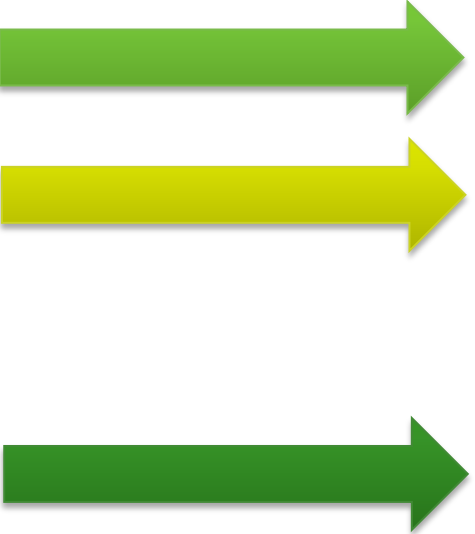
New Programming Model



```
class LogicalSwitch {  
    boolean enabled;  
    boolean enablePort(boolean enable) {  
        boolean wasEnabled = enabled;  
        enabled = enable;  
        return wasEnabled;  
    }  
}
```

New Programm

```
class LogicalSwitch {  
    boolean enabled;  
  
    boolean enablePort(boolean enable)  
        boolean wasEnabled = enabled;  
        enabled = enable;  
        return wasEnabled;  
}
```



```
CorfuRuntime cr =  
    new CorfuRuntime("10.0.0.1:9000");  
  
LogicalSwitch mySwitch =  
    cr.build()  
        .setStreamName("switch-a")  
        .setType(LogicalSwitch.class)  
        .open();  
  
mySwitch.enablePort(true);
```


New Programm

```
class LogicalSwitch {
```

```
    boolean enabled;
```

```
    boolean enablePort(boolean enable)  
    {  
        boolean wasEnabled = enabled;  
        enabled = enable;  
        return wasEnabled;  
    }
```

```
}
```

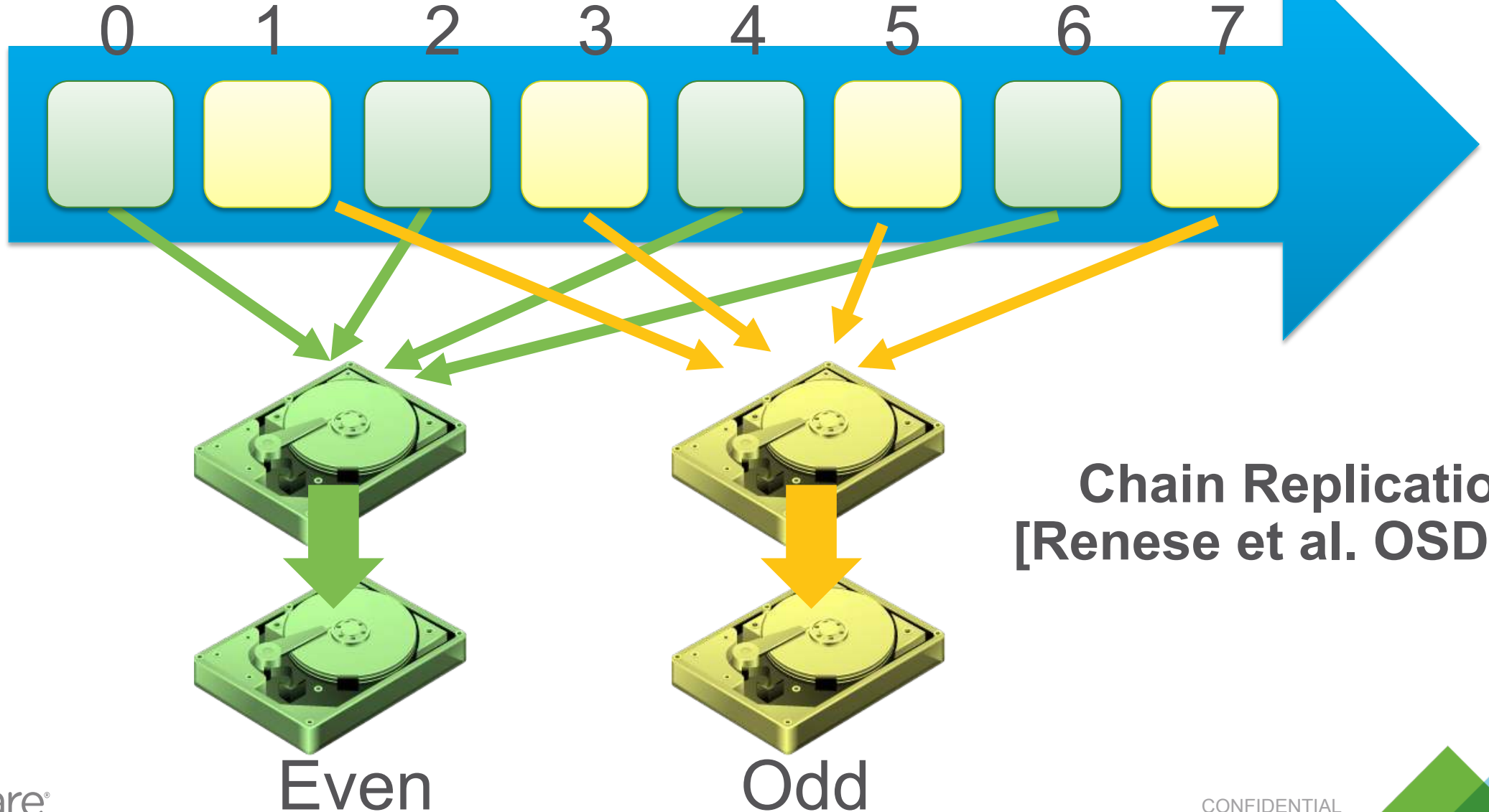
```
CorfuRuntime cr =  
    new CorfuRuntime("localhost:9000");
```

```
LogicalSwitch mySwitch =  
    cr.build()  
        .setStreamName("switch-a")  
        .setType(LogicalSwitch.class)  
        .open();
```

```
mySwitch.enablePort(true);
```

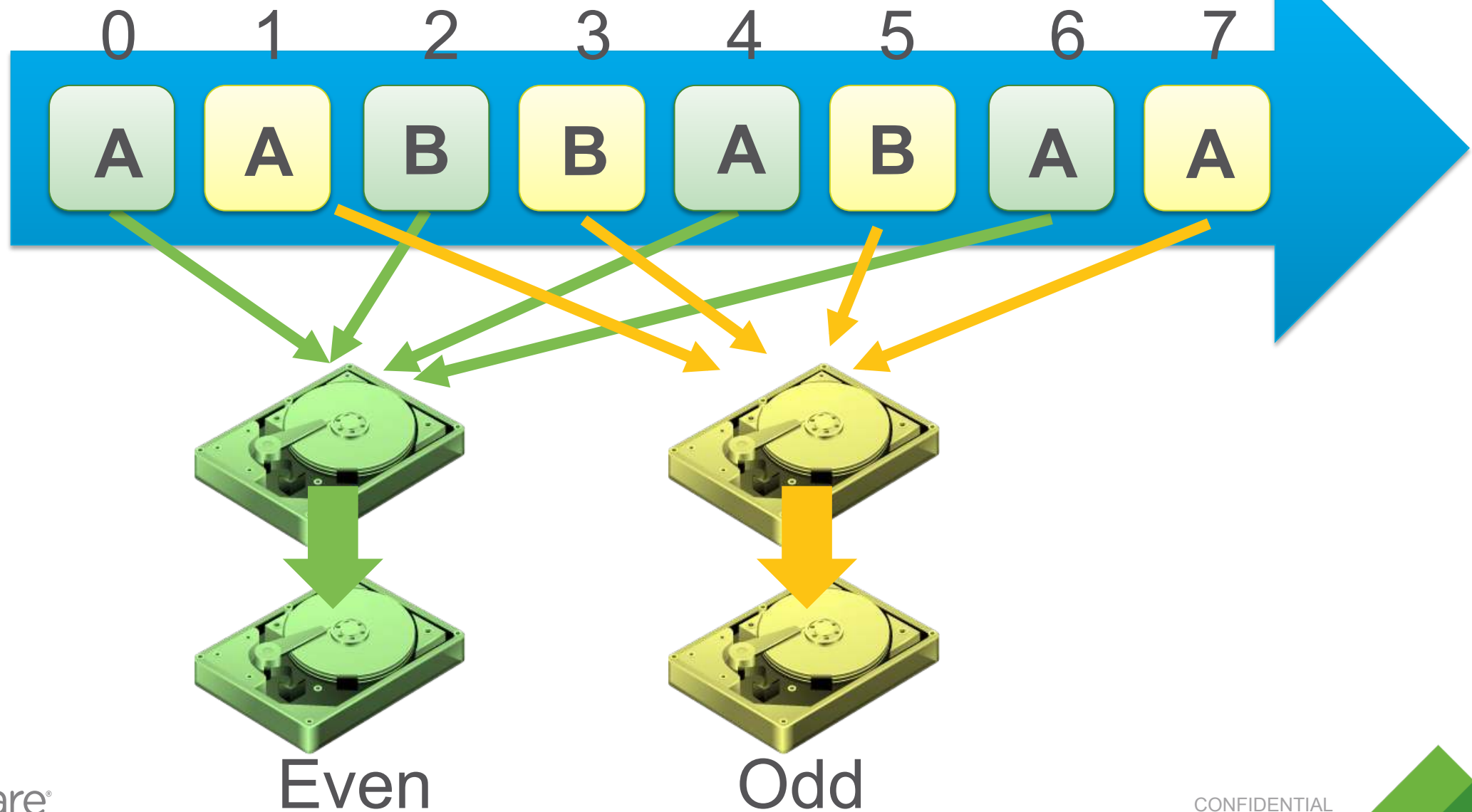


Supporting Multiple Objects

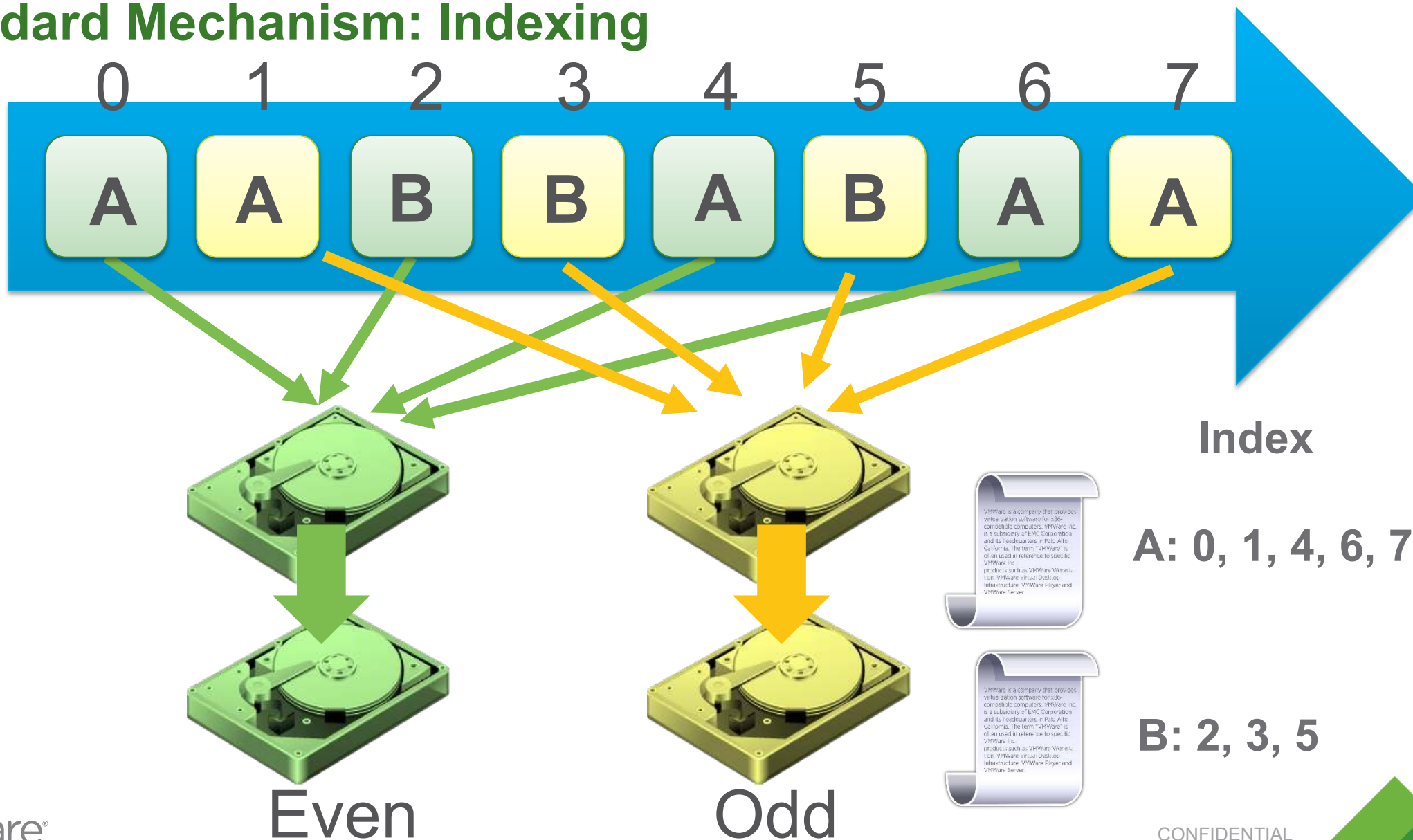


Chain Replication
[Renese et al. OSDI'04]

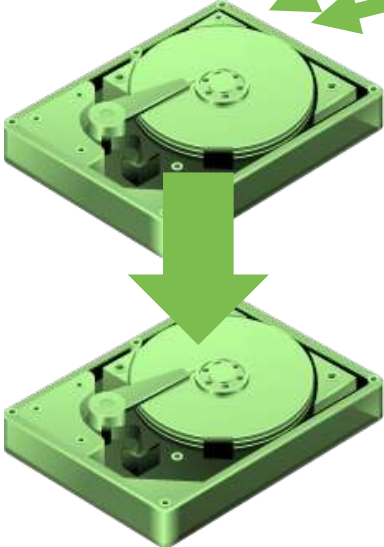
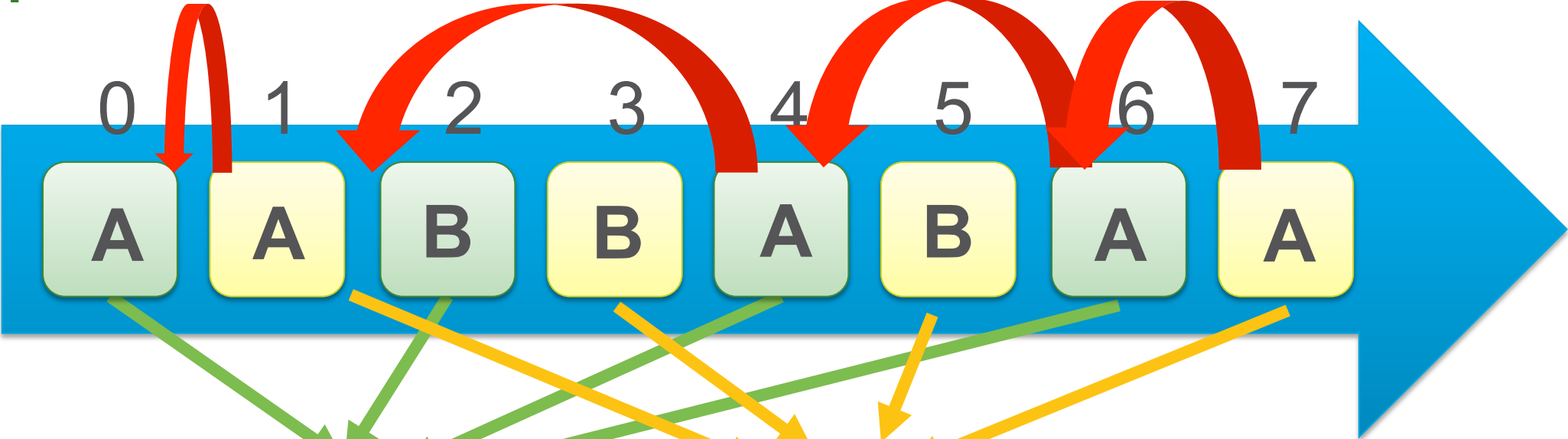
Supporting Multiple Objects



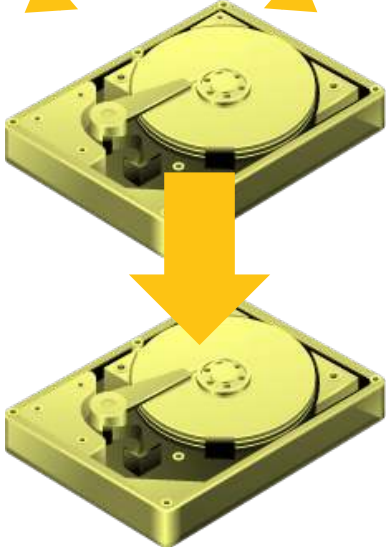
Standard Mechanism: Indexing



Backpointers



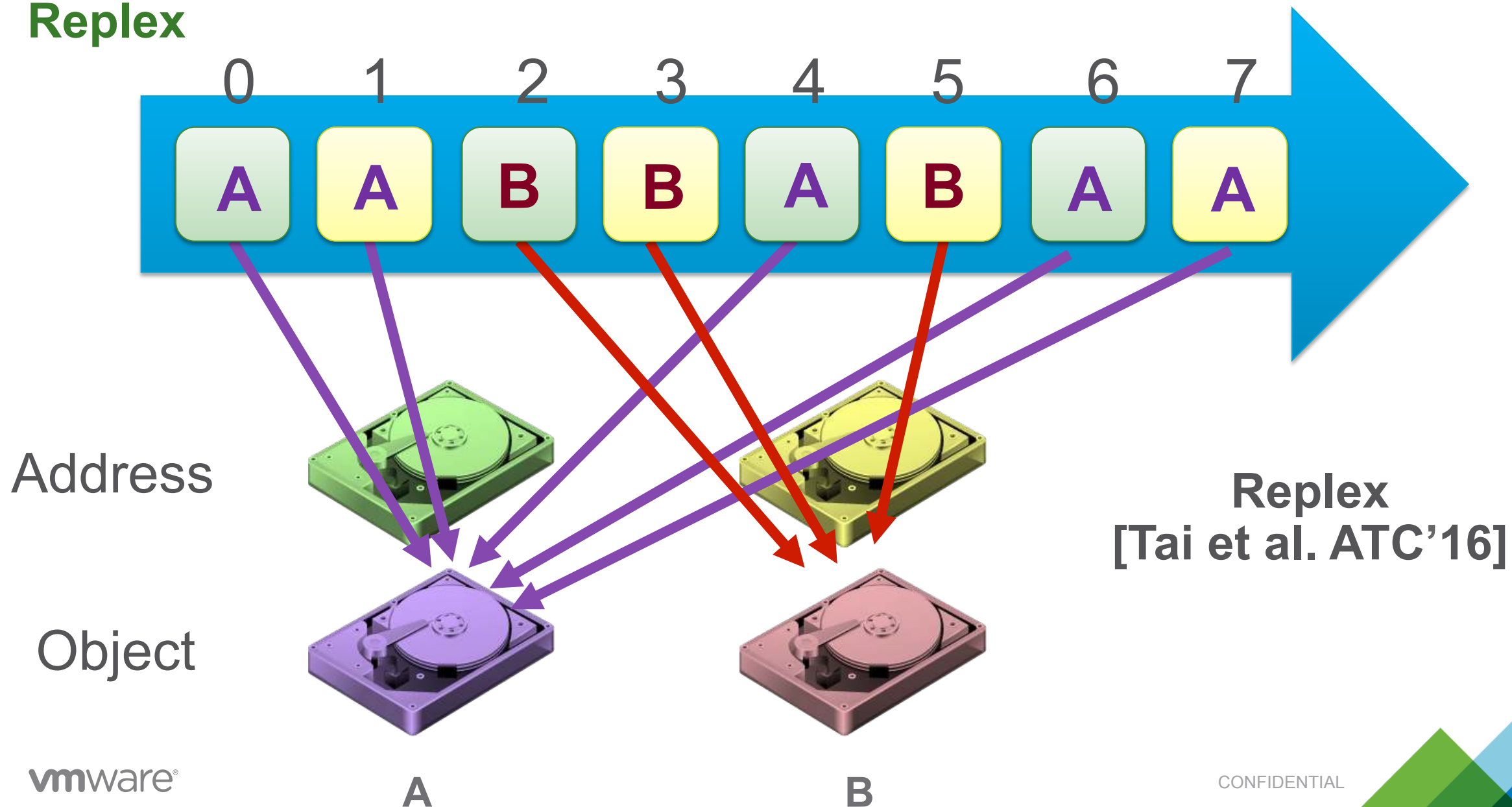
Even



Odd

Tango
[Balakrishnan et al.
SOSP '13]

Replex

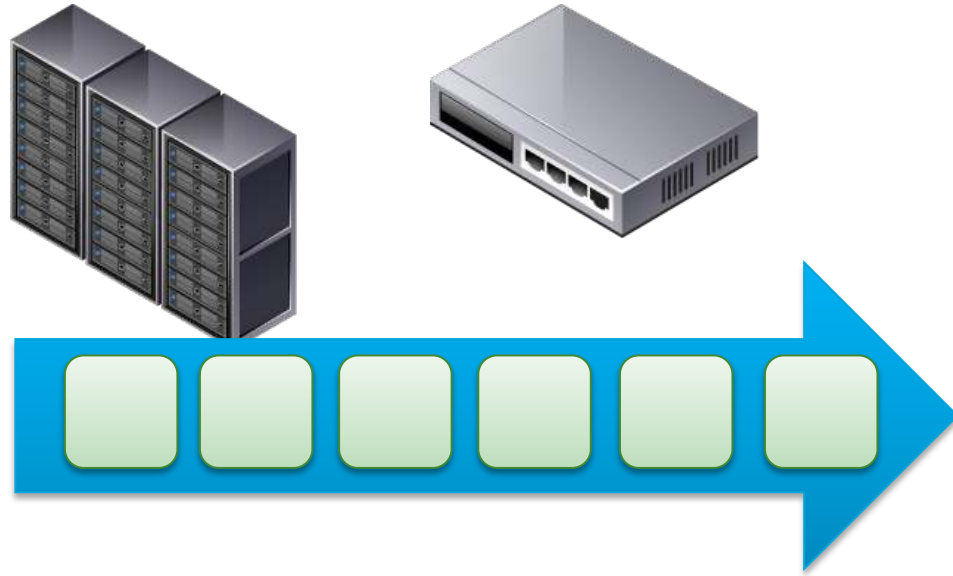


CMPLAT: Cluster Management Platform for NSX

Left to Right:
Scott Fritchie
Michael Wei(PoC),
Chris Rossbach,
Dahlia Malkhi (PoC),
Udi Wieder



Jim Stabile, Ittai Abraham,
Sandeep Uttamchandani
Medhavi Dawan,
Maithem Munshed



- Re-engineer NSX cluster management to use Corfu “under the covers”
- Avoid the need for every NSX installation to include Zookeeper, etc.
- Corfu evolved significantly as a result of this joint effort

CMPLAT: Cluster Management Platform for NSX

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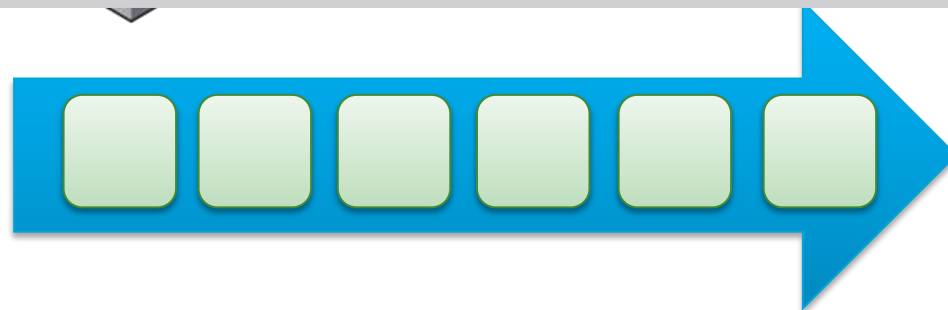
API Requests



NSX Scale-Out Management Plane

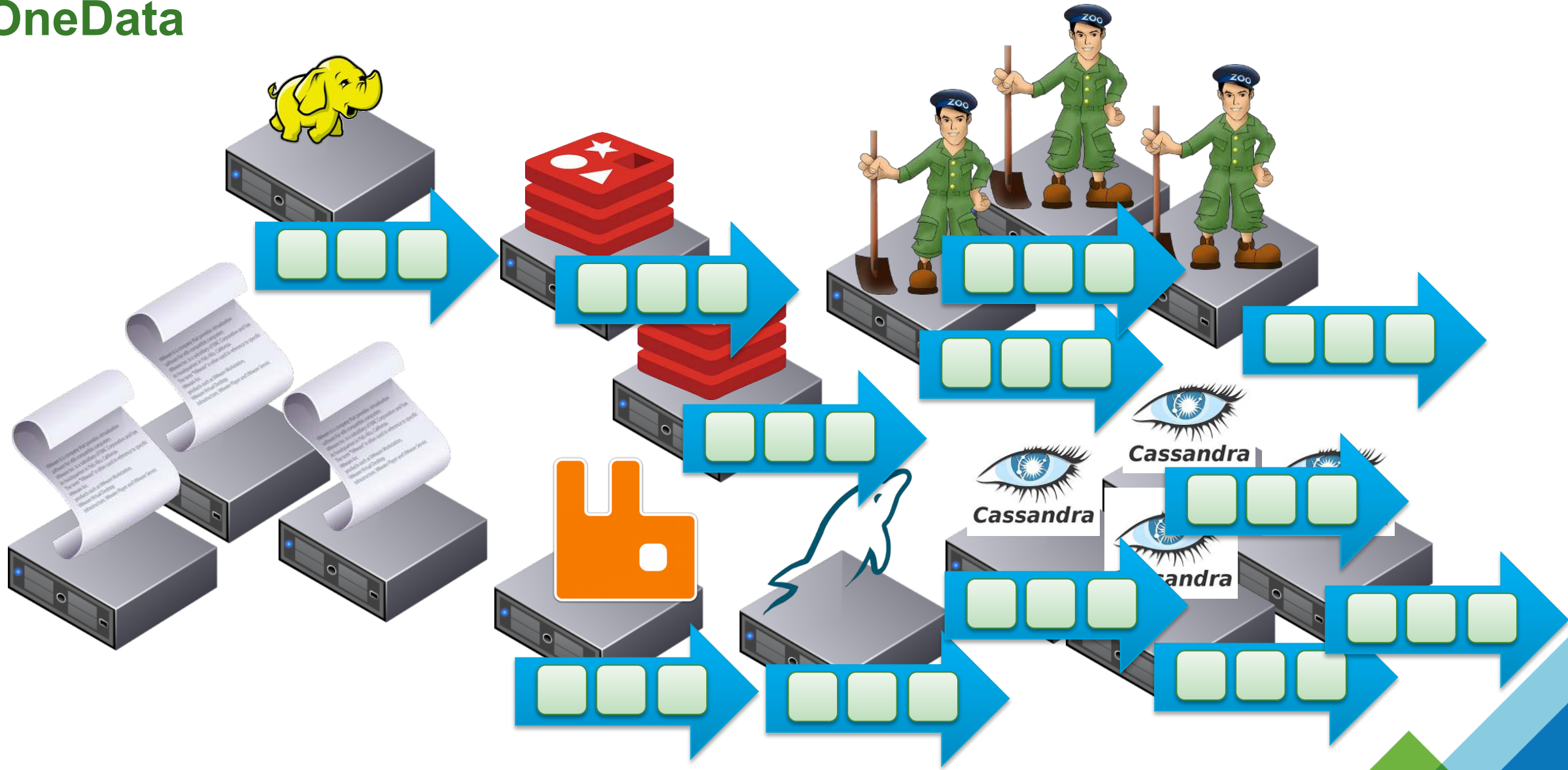


NSX Java Object Model



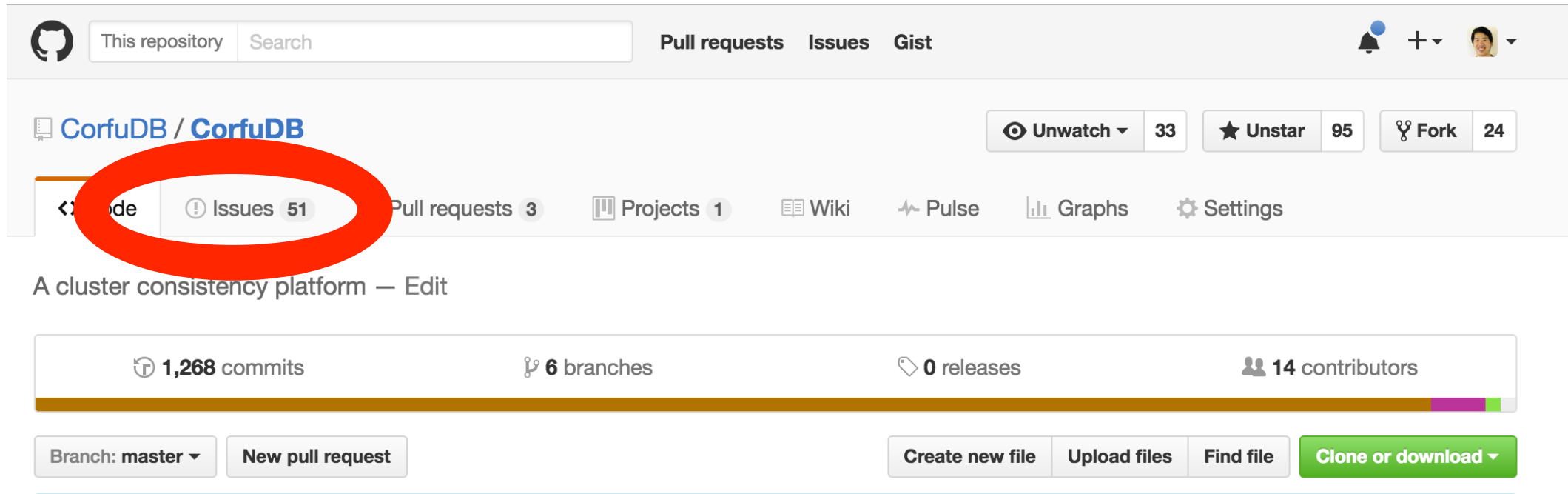
Corfu Distributed Log

OneData



Open Source Efforts

Yes, we're on Github...
github.com/CorfuDB/CorfuDB



No, this is not a source code dump...